

A: Balanced System Reactor

Source file name: `balsys.c`, `balsys.cpp`, `balsys.java`, or `balsys.py`

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In an experimental industrial facility, engineers work with chains of modular energy reactors. Each reactor module is assigned a positive integer power factor that determines its individual contribution to the system.

The *total linear output* of the system is the sum of all individual contributions from the modular energy reactors. The *compound amplification* produced by the system can be calculated as the product of all individual power factors from the modular energy reactors.

Due to strict safety regulations, the system is considered *perfectly balanced* only if the following condition is satisfied:

The total linear output of the system is exactly equal to the compound amplification produced by chaining all reactor modules together.

Formally, if the power factors of the n reactor modules are $x_1, x_2, \dots, x_n$, the system is perfectly balanced if and only if

$$x_1 + \dots + x_n = x_1 \cdot \dots \cdot x_n.$$

At the very beginning of the project the facility was very small. There were only 3 reactors with powerfactors 1, 2, 3. Clearly, the system was balanced, since $1 + 2 + 3 = 1 \cdot 2 \cdot 3$.

Now the project is mature enough to think in adding some more reactors in order to produce a greater total linear output. Anyway, engineers know that they want a balanced system, and they find that using several reactors with power factor equal to 1, they could achieve a balanced system. For instance, they could have 8 reactors with power factors 1, 1, 1, 1, 1, 1, 2, 8, the total linear output and the compound amplification are both 16.

However, safety analysts have also determined that many reactors with power factor 1 can lead to resonance instabilities, with eventual dangerous consequences. For this case, they prefer to have a power factor configuration like 1, 1, 1, 1, 1, 2, 2, 3. It is balanced and has only 5 reactors with power factor 1. Therefore, they require the following additional restriction:

For $n > 3$, exactly $n - 3$ reactor modules must share power factor 1.

Enforcing this restriction may mean that there is no way to balance the system, for some values of the system size n (the number of reactor modules). Anyway, given n , your task is to decide if configuring the system in a balanced way is possible and, if this is the case, tell how to do it.

Input

There are several cases to solve. Each case is defined with just a line with a single integer n , the number of reactor modules in the system ($4 \leq n \leq 10^7$). The end of the input is signaled with a line containing 0.

The input must be read from standard input.

Output

For each case, print a line with the values of the power factors different from 1, in ascending order, if the system may be configured in a balanced way. Otherwise, print IMPOSSIBLE.

The output must be written to standard output.

Sample Input	Sample Output
8	2 2 3
13	2 3 3
6	IMPOSSIBLE
0	